Trustworthy Dataset Distillation via Variance-Aware Crop Selection

A trustworthiness study of RDED

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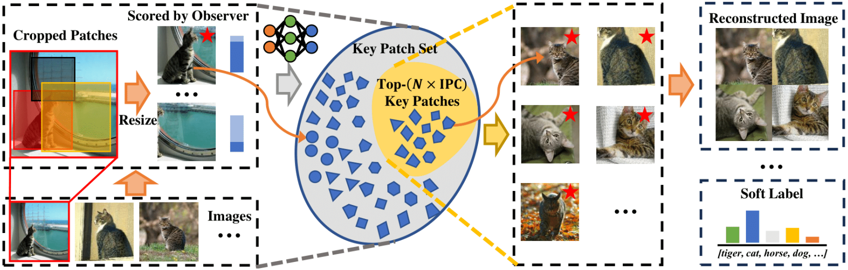
June 23, 2026

Outline

RDED: optimization-free dataset distillation [1]



Pipeline (per class): crop real images $\rightarrow$ score each crop with a *frozen* teacher ($\ell_j = -\log \text{softmax}(z_j)_{[y]}$) $\rightarrow$ keep the most-confident crops $\rightarrow$ stitch into IPC images/class $\rightarrow$ **relabel** with the teacher's soft labels. A **student** is trained *only* on this set by knowledge distillation.

Diagnostic overview — student vs. teacher (full grid)

		top-1		ECE		AUROC		OSCR	
cell	**IPC**	S	T	S	T	S	T	S	T
C100/Conv3	1	21.6	59.7	.203	.104	.647	.946	.169	.577
	10	48.6	59.7	.097	.104	.903	.946	.457	.577
	50	57.5	59.7	.097	.104	.928	.946	.549	.577
C100/RN18	1	11.2	70.9	.274	.122	**.379**	.946	.059	.690
	10	44.7	70.9	.190	.122	.765	.946	.391	.690
	50	63.3	70.9	.132	.122	.894	.946	.594	.690
Tiny/Conv4	1	13.3	50.2	.177	.065	**.423**	.982	.080	.497
	10	39.9	50.2	.064	.065	.626	.982	.305	.497
	50	48.3	50.2	.056	.065	.915	.982	.464	.497
Tiny/RN18	1	8.9	62.0	.333	.126	**.193**	.981	.024	.616
	10	43.7	62.0	.154	.126	.706	.981	.373	.616
	50	58.2	62.0	.115	.126	.905	.981	.559	.616

S = student (trained on RDED), **T** = teacher (real-data ceiling); ID = real val, OOD = SVHN (energy score).
Good: top-1/AUROC/OSCR $\uparrow$, ECE $\downarrow$. [tools/show_diagnostics.py]

H1 — the distilled set is over-collapsed (all IPCs)

Teacher-induced NC1 $= \frac{1}{K}\,\mathrm{tr}(\Sigma_W \Sigma_B^{+})$, [2] — lower = more collapsed

cell	**IPC**	**distilled NC1**	**real-subset NC1**	**ratio**	**teacher NC1**
C100/Conv3	1	0.00	0.00	–	8.13
	10	0.89	1.52	0.59	8.13
	50	2.70	5.15	0.52	8.13
C100/RN18	1	0.00	0.00	–	6.62
	10	0.95	1.91	0.49	6.62
	50	1.73	2.91	0.59	6.62
Tiny/Conv4	1	0.00	0.00	–	8.97
	10	1.56	2.21	0.71	8.97
	50	5.18	8.67	0.60	8.97
Tiny/RN18	1	0.00	0.00	–	15.19
	10	1.88	4.42	0.43	15.19
	50	3.88	7.71	0.50	15.19

H2 — the trained student, in isolation (full grid)

Student on real val (mean / 3 seeds); OOD = SVHN, energy score. NC1 here is the *student's* geometry on real val.

cell	IPC	top-1	ECE	OSCR	AUROC	FPR95	NC1
C100/Conv3	1	21.6	.203	.169	.647	.934	15.6
	10	48.6	.097	.457	.903	.381	9.2
	50	57.5	.097	.549	.928	.287	8.4
C100/RN18	1	11.2	.274	.059	**.379**	.991	34.8
	10	44.7	.190	.391	.765	.820	12.7
	50	63.3	.132	.594	.894	.531	7.8
Tiny/Conv4	1	13.3	.177	.080	**.423**	.994	17.2
	10	39.9	.064	.305	.626	.939	12.3
	50	48.3	.056	.464	.915	.452	9.7
Tiny/RN18	1	8.9	.333	.024	**.193**	1.000	45.7
	10	43.7	.154	.373	.706	.994	21.0
	50	58.2	.115	.559	.905	.587	14.3

Selectors: only the crop-selection rule changes

For one class, candidate crops $\{x_j\}$ have teacher features ϕ_j and realism $\ell_j = -\log \mathrm{softmax}(z_j)_{[y]}$. Each method picks a size-n set S. **Relabeling and KD training are identical to stock** — so any trust change is *causal*.

stock (RDED): the n smallest-ℓ_j crops. *Most-confident crops sit near the class mean* $\Rightarrow \Sigma_W$ biased small $\Rightarrow$ over-collapse.

Realism floor (shared guard): pool $P =$ top $\lceil r\, n \rceil$ most-realistic crops ($r = 3$ default); all variance-seekers choose $S \subseteq P$.

random: uniform over P (weak lever).

covmatch: maximize feature **volume**

$$S = \arg\max_{S \subseteq P} \log\det\big(K_S + \epsilon I\big)$$

on the cosine kernel $K_{ij} = \langle \tilde{\phi}_i, \tilde{\phi}_j \rangle$ (greedy DPP-MAP [3]) — injects within-class spread.

Dose-response — energy-AUROC, every IPC & cell

cell	IPC	stock	random	covmatch	Δ(cov−stock)
C100/Conv3	1	.725	.690	.684	−.041
	10	.892	.913	**.921**	+.029
	50	.930	.938	.931	+.001
C100/RN18	1	.443	.411	.449	+.006
	10	.751	.771	**.786**	+.035
	50	.879	.892	**.901**	+.022
Tiny/Conv4	1	.438	.481	.427	−.011
	10	.707	.700	**.742**	+.035
	50	.915	.928	**.955**	+.040
Tiny/RN18	1	.288	.236	.265	−.023
	10	.657	.692	.684	+.027
	50	.852	.901	.895	+.043

covmatch lifts the energy-AUROC over stock at **IPC $\geq$ 10** in **all four cells** ($\Delta > 0$); on this logit-based score the `random` midpoint is noisier than on the geometry-aware Mahalanobis (and in a few cells `random` edges out `covmatch`). **IPC 1: no gain / regresses** (no within-class variance to restore). [`analyze_select.py`]

References I

[1] Peng Sun, Bei Shi, Daiwei Yu, and Tao Lin. On the diversity and realism of distilled dataset: An efficient dataset distillation paradigm. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[2] Vardan Papyan, X. Y. Han, and David L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences*, 117(40):24652–24663, 2020.

[3] Laming Chen, Guoxin Zhang, and Eric Zhou. Fast greedy map inference for determinantal point process to improve recommendation diversity. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

Thank you!